Dini's Theorem

We will end this section with a famous theorem due to Ulisse Dini (1845–1918) that gives a partial converse to Theorem 8.2.2 when the sequence is monotone. We will present a proof using nonconstant gauges (see Section 5.5).

8.2.6 Dini's Theorem *Suppose that (f_n) is a monotone sequence of continuous functions on $I := [a, b]$ that converges on I to a continuous function f. Then the convergence of the sequence is uniform.*

Proof. We suppose that the sequence (f_n) is decreasing and let $g_m := f_m - f$. Then (g_m) is a decreasing sequence of continuous functions converging on I to the 0-function. We will show that the convergence is uniform on I.

Given $\varepsilon > 0$, $t \in I$, there exists $m_{\varepsilon,t} \in \mathbb{N}$ such that $0 \leq g_{m_{\varepsilon,t}}(t) < \varepsilon/2$. Since $g_{m_{\varepsilon,t}}$ is continuous at t, there exists $\delta_\varepsilon(t) > 0$ such that $0 \leq g_{m_{\varepsilon,t}}(x) < \varepsilon$ for all $x \in I$ satisfying $|x - 1| \leq \delta_\varepsilon(t)$. Thus, δ_ε is a gauge on I, and if $\dot{\mathcal{P}} = \{(I_i, t_i)\}_{i=1}^n$ is a δ_ε-fine partition, we set $M_\varepsilon := \max\{m_{\varepsilon,t_1}, \ldots, m_{\varepsilon,t_n}\}$. If $m \geq M_\varepsilon$ and $x \in I$, then (by Lemma 5.5.3) there exists an index i with $|x - t_i| \leq \delta_\varepsilon(t_i)$ and hence

$$0 \leq g_m(x) \leq g_{m,t_i}(x) < \varepsilon.$$

Therefore, the sequence (g_m) converges uniformly to the 0-function. Q.E.D.

It will be seen in the exercises that we cannot drop any one of the three hypotheses: (i) the functions f_n are continuous, (ii) the limit function f is continuous, (iii) I is a closed bounded interval.

Exercises for Section 8.2

1. Show that the sequence $(x^n/(1 + x^n))$ does not converge uniformly on $[0, 2]$ by showing that the limit function is not continuous on $[0, 2]$.
2. Prove that the sequence in Example 8.2.1(c) is an example of a sequence of continuous functions that converges nonuniformly to a continuous limit.
3. Construct a sequence of functions on $[0, 1]$ each of which is discontinuous at every point of $[0, 1]$ and which converges uniformly to a function that is continuous at every point.
4. Suppose (f_n) is a sequence of continuous functions on an interval I that converges uniformly on I to a function f. If $(x_n) \subseteq I$ converges to $x_0 \in I$, show that $\lim(f_n(x_n)) = f(x_0)$.
5. Let $f : \mathbb{R} \to \mathbb{R}$ be uniformly continuous on $\mathbb{R}$ and let $f_n(x) := f(x + 1/n)$ for $x \in \mathbb{R}$. Show that (f_n) converges uniformly on $\mathbb{R}$ to f.
6. Let $f_n(x) := 1/(1 + x)^n$ for $x \in [0, 1]$. Find the pointwise limit f of the sequence (f_n) on $[0, 1]$. Does (f_n) converge uniformly to f on $[0, 1]$?
7. Suppose the sequence (f_n) converges uniformly to f on the set A, and suppose that each f_n is bounded on A. (That is, for each n there is a constant M_n such that $|f_n(x)| \leq M_n$ for all $x \in A$.) Show that the function f is bounded on A.
8. Let $f_n(x) := nx/(1 + nx^2)$ for $x \in A := [0, \infty)$. Show that each f_n is bounded on A, but the pointwise limit f of the sequence is not bounded on A. Does (f_n) converge uniformly to f on A?

9. Let $f_n(x) := x^n/n$ for $x \in [0, 1]$. Show that the sequence (f_n) of differentiable functions converges uniformly to a differentiable function f on $[0, 1]$, and that the sequence (f_n') converges on $[0, 1]$ to a function g, but that $g(1) \neq f'(1)$.
10. Let $g_n(x) := e^{-nx}/n$ for $x \geq 0, n \in \mathbb{N}$. Examine the relation between $\lim(g_n)$ and $\lim(g_n')$.
11. Let $I := [a, b]$ and let (f_n) be a sequence of functions on $I \to \mathbb{R}$ that converges on I to f. Suppose that each derivative f_n' is continuous on I and that the sequence (f_n') is uniformly convergent to g on I. Prove that $f(x) - f(a) = \int_a^x g(t)dt$ and that $f'(x) = g(x)$ for all $x \in I$.
12. Show that $\lim \int_1^2 e^{-nx^2} dx = 0$.
13. If $a > 0$, show that $\lim \int_a^\pi (\sin nx)/(nx)dx = 0$. What happens if $a = 0$?
14. Let $f_n(x) := nx/(1 + nx)$ for $x \in [0, 1]$. Show that (f_n) converges nonuniformly to an integrable function f and that $\int_0^1 f(x)dx = \lim \int_0^1 f_n(x)dx$.
15. Let $g_n(x) := nx(1 - x)^n$ for $x \in [0, 1]$, $n \in \mathbb{N}$. Discuss the convergence of (g_n) and $(\int_0^1 g_n dx)$.
16. Let $\{r_1, r_2, \ldots, r_n \ldots\}$ be an enumeration of the rational numbers in $I := [0, 1]$, and let $f_n : I \to \mathbb{R}$ be defined to be 1 if $x = r_1, \ldots, r_n$ and equal to 0 otherwise. Show that f_n is Riemann integrable for each $n \in \mathbb{N}$, that $f_1(x) \leq f_2(x) \leq \cdots \leq f_n(x) \leq \cdots$, and that $f(x) := \lim(f_n(x))$ is the Dirichlet function, which is not Riemann integrable on $[0, 1]$.
17. Let $f_n(x) := 1$ for $x \in (0, 1/n)$ and $f_n(x) := 0$ elsewhere in $[0, 1]$. Show that (f_n) is a decreasing sequence of discontinuous functions that converges to a continuous limit function, but the convergence is not uniform on $[0, 1]$.
18. Let $f_n(x) := x^n$ for $x \in [0, 1]$, $n \in \mathbb{N}$. Show that (f_n) is a decreasing sequence of continuous functions that converges to a function that is not continuous, but the convergence is not uniform on $[0, 1]$.
19. Let $f_n(x) := x/n$ for $x \in [0, \infty)$, $n \in \mathbb{N}$. Show that (f_n) is a decreasing sequence of continuous functions that converges to a continuous limit function, but the convergence is not uniform on $[0, \infty)$.
20. Give an example of a decreasing sequence (f_n) of continuous functions on $[0, 1)$ that converges to a continuous limit function, but the convergence is not uniform on $[0, 1)$.

Section 8.3 The Exponential and Logarithmic Functions

We will now introduce the exponential and logarithmic functions and will derive some of their most important properties. In earlier sections of this book we assumed some familiarity with these functions for the purpose of discussing examples. However, it is necessary at some point to place these important functions on a firm foundation in order to establish their existence and determine their basic properties. We will do that here. There are several alternative approaches one can take to accomplish this goal. We will proceed by first proving the existence of a function that has *itself* as derivative. From this basic result, we obtain the main properties of the exponential function. The logarithm function is then introduced as the inverse of the exponential function, and this inverse relation is used to derive the properties of the logarithm function.

The Exponential Function

We begin by establishing the key existence result for the exponential function.

8.3.1 Theorem *There exists a function $E : \mathbb{R} \to \mathbb{R}$ such that:*

(i) $E'(x) = E(x)$ *for all* $x \in \mathbb{R}$.

(ii) $E(0) = 1$.